INCH-POUND

MIL-DTL-27072/42C
12 March 2012
SUPERSEDING
MIL-DTL-27072/42B
8 December 2000

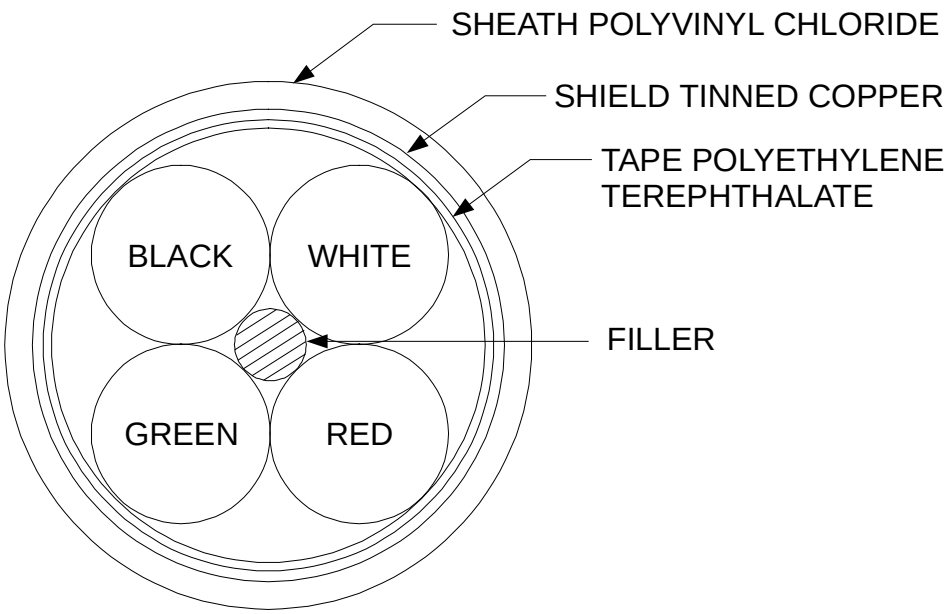
DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR (4 CONDUCTORS) AND SINGLE SHIELDED, M27072/42JBCA*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-27072.

REQUIREMENTS:




NOTE: The complete Part or Identifying Number (PIN) requires a color code designator (see table I).

FIGURE 1. Cable configuration.

AMSC N/A FSC 6145

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| | |
|---|---|
| Design and construction: | See figure 1 |
| Number of conductors: | 4 |
| Temperature rating: | 105 °C |
| Components: | |
| Basic wire: | M16878/18BGE*** (see table I) |
| Conductor: | Tinned copper |
| Primary insulation: | Polyvinyl chloride |
| Colors: | Black<br>White<br>Red<br>Green |
| Jacket over insulation: | Polyamide |
| Color: | Natural |
| Filler: | |
| Material: | In accordance with MIL-DTL-27072 |
| Lay: | 1.8 inches maximum |
| Cabling: | Not applicable |
| Binder tape material: | Polyethylene terephthalate |
| Shield: | Tinned copper |
| Gauge: | 36 AWG |
| Braid angle: | 20 to 40 degrees |
| Jacket undershield: | Not applicable |
| Sheath: | |
| Material: | Polyvinyl chloride |
| Wall thickness: | .025 inch nominal |
| Tensile strength: | In accordance with MIL-DTL-27072 |
| Ultimate elongation: | In accordance with MIL-DTL-27072 |
| Accelerated aging: | In accordance with MIL-DTL-27072 |
| Color: | See MIL-DTL-27072 |
| Finished cable: | See table I |
| Cold bend: | Mandrel size in accordance with MIL-DTL-27072 |
| Dielectric withstanding voltage: | 3,000 V rms |
| Conductor resistance: | In accordance with basic wire specification |

TABLE I. Construction data.

| PIN 27072/42 | Basic wire except color M16878/18BGE***1/,2/ (reference) | | | | Finished cable diameter inch(mm) ±10% |
|---|---|---|---|---|---|
| | Wire size (AWG) | Number of strands | Strand size | Nominal diameter inch(mm) | |
| JBCA* | 20 | 19 | 32 | .078(1.98) | .270(6.02) |

1/ Asterisks (*) shall be replaced by the appropriate color designator.
2/ Dimensions are in inches. Metric equivalents are given for information only.



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Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Referenced document. This document references MIL-DTL-27072.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
DLA – CC

Preparing activity:
DLA - CC

(Project 6145-2012-023)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.daps.dla.mil.



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